

KRITAGYA YADAV

Java Backend Developer — Software Engineer

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Professional Summary

- B.Tech Computer Science student (Graduating 2027, CGPA 7.58) with hands-on experience across the full Software Development Life Cycle (SDLC) – from database design and implementation through automated testing and CI/CD deployment. Independently designed, built, and deployed 83+ RESTful APIs across two full-stack Java/Spring Boot systems, including a live cloud-deployed backend on Render, applying strong data structures, algorithms, and object-oriented design fundamentals – validated by 55+ automated JUnit/Mockito unit and integration tests. Comfortable working independently, solving problems end-to-end, and quickly picking up new tools and technologies.

Technical Skills

- **Languages:** Java, SQL, JavaScript, HTML, CSS
- **Backend & API Development:** Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate (ORM), RESTful API Design, WebSocket (STOMP)
- **Databases & SQL:** PostgreSQL, MySQL, Simple & Complex SQL Queries, Joins, Indexing, Redis (Caching), Database/Schema Design
- **Authentication & Security:** JWT Authentication, Refresh Token Rotation, Role-Based Access Control (RBAC), BCrypt Password Encryption
- **Cloud & DevOps:** Docker, Docker Compose, Multi-Stage Builds, GitHub Actions (CI/CD), Render (Cloud Deployment), Nginx
- **Testing:** JUnit 5, Mockito, Unit Testing, Integration Testing, Bean Validation
- **Tools:** Git, GitHub, Version Control, Maven, IntelliJ IDEA, Postman, Swagger/OpenAPI (API Documentation), Linux CLI
- **Core CS & Design Concepts:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Software Development Life Cycle (SDLC), System Design, API Design, MVC Architecture, Dependency Injection, Concurrency Control, DBMS, Operating Systems, Computer Networks

Projects

LinkPulse – URL Shortener Platform | *Java, Spring Boot, PostgreSQL, Docker, Nginx, Render, GitHub Actions*
GitHub | **Live Demo**

- Deployed a production Java/Spring Boot backend to the cloud (Render) with a fully automated CI/CD pipeline: GitHub Actions builds and runs the JUnit test suite, pushes a Docker image to Docker Hub, then triggers an automatic Render deployment on every merge to main.
- Designed and implemented a RESTful API with 5 endpoints using a layered Controller-Service-Repository (MVC) architecture, featuring a Base62 short-code redirect engine with indexed database lookups and HikariCP connection pooling for performance.
- Verified reliability with 16 automated JUnit unit tests integrated directly into the CI/CD pipeline, configured to block Docker image builds on any failing test.
- Reduced production debugging time by centralizing exception handling via Spring's `@RestControllerAdvice`, standardizing JSON error responses across the REST API.
- Built a live analytics dashboard (JavaScript, HTML5, CSS3) consuming the REST API, with auto-polling click analytics, search/filtering, status badges, automated link expiration (HTTP 410 Gone), and server-side Bean Validation.

Event Booking System | *Java, Spring Boot, PostgreSQL, Redis, React, WebSocket*

GitHub

- Designed and engineered 83 RESTful API endpoints across 16 controllers and 13 JPA entities, applying layered MVC architecture, Dependency Injection, and relational database design to support authentication, booking, payments, and admin workflows.
- Solved seat-overbooking race conditions under concurrent load by implementing optimistic locking (`@Version`) and concurrency control on Seat, Booking, and Event entities, combined with automated seat-lock expiration scheduling.
- Delivered real-time seat availability to concurrent users by building a WebSocket (STOMP) messaging layer on top of Redis caching, eliminating stale-availability bugs from the booking flow.
- Secured all 83 API endpoints by implementing JWT authentication with refresh token rotation, Role-Based Access Control (RBAC), and BCrypt password encryption across the application.
- Validated system correctness with 39 automated JUnit/Mockito unit and integration tests, including a dedicated `SecurityFixesTest` regression suite, and shipped QR-code e-ticketing, coupons, and waitlists, fully documented via Swagger/OpenAPI.

Education

KCC Institute of Technology & Management, AKTU

2023 – 2027

Bachelor of Technology (B.Tech) – Computer Science & Engineering

CGPA: 7.58/10